Suppression of Cascading Failure in the IEEE-39 System via Logistic Softening with Memory Feedback

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We present a proof-of-concept study of adaptive control in a benchmark power grid under stochastic loading. Using the IEEE-39 system, we compare an uncontrolled baseline with a control mechanism based on logistic softening and memory-dependent feedback. Monte Carlo simulations reveal strong suppression of cascading failure in a near-critical regime. Across a load sweep, the controlled system exhibits a markedly reduced collapse probability relative to the baseline over the intermediate stress range. At a representative operating point of 140 MW, collapse probability is reduced from approximately 0.56 in the baseline system to approximately 0.01 under control. These results demonstrate that the proposed mechanism can stabilize grid dynamics in a standard benchmark system and provide a foundation for extension to larger networks exhibiting richer critical structure.

I. INTRODUCTION

Cascading failure in electric power systems remains a central problem in the study of complex infrastructure networks. Under increasing stress, local overloads and voltage violations may propagate through the system, leading to large-scale failure events. Classical approaches often treat such transitions as sharp critical points; however, adaptive control mechanisms may significantly alter this behavior.

In this work, we study a compact benchmark system, IEEE-39, as a controlled test environment for a stabilization mechanism based on logistic softening combined with memory feedback. The goal is not to establish universality at this stage, but to demonstrate clearly that the mechanism produces a strong and measurable suppression effect in a standard power-grid model.

The IEEE-39 system is particularly suitable for this purpose because of its moderate size and well-understood structure. It allows us to isolate the effect of the control mechanism without the additional complexity introduced by large-scale heterogeneity.

II. METHODS

A. Simulation framework

Simulations were performed on the IEEE-39 benchmark power network using repeated Monte Carlo trials. At each operating point, we compare:

- an uncontrolled baseline system,
- a controlled system with logistic softening and memory feedback.

Each trial evolves for a fixed number of time steps under stochastic demand perturbations modeled as Gaussian noise.

B. Control mechanism

The control mechanism combines logistic softening with memory feedback. Let $x(t)$ denote a scalar stress indicator derived from system loading conditions, constructed from aggregate load-generation imbalance and network stress metrics (e.g., line loading or voltage deviation). The control response is given by

$$u(t) = \frac{1}{1 + \exp\left[\beta\left(m(t) - m_c\right)\right]}, \tag{1}$$

where the memory variable evolves as

$$m(t) = \alpha m(t-1) + (1-\alpha)x(t). \tag{2}$$

The memory variable is initialized as $m(0) = x(0)$.

Here, β controls the sharpness of the response, m_c is a critical threshold, and α determines the strength of memory feedback.

C. Collapse diagnostics

Collapse events are defined using operational criteria such as line overload, transformer overload, and minimum voltage threshold violation. The primary observable is collapse probability:

$$P_{\text{collapse}} = \frac{N_{\text{collapse}}}{N_{\text{runs}}}. \tag{3}$$

D. Parameter settings

In the simulations presented here, the control parameters are set to $\beta = 9.0$, $m_c = 1.05$, and $\alpha = 0.6$.

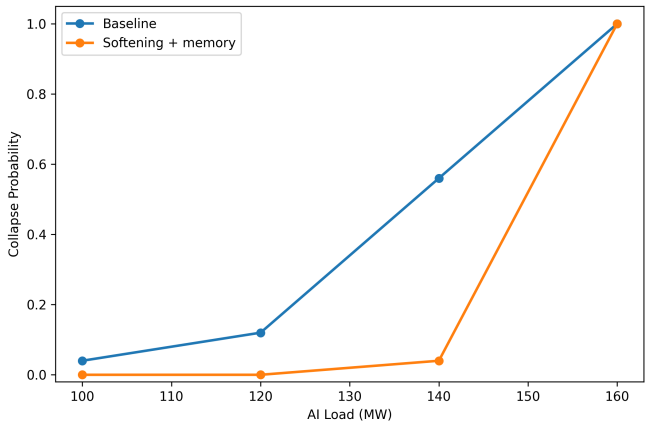

Figure 1. Collapse probability as a function of injected load in the IEEE-39 system. The baseline exhibits rapid failure growth, while the controlled system remains strongly suppressed in the near-critical regime.

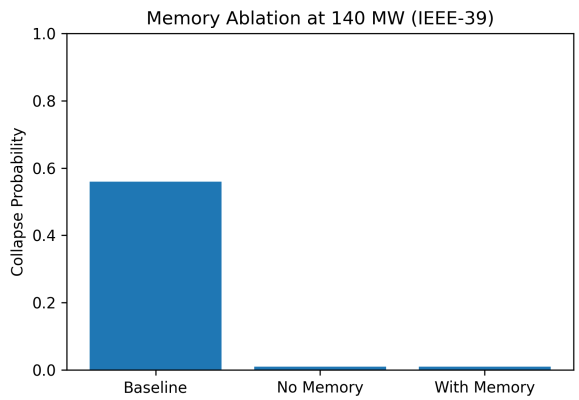

Figure 2. Memory ablation at 140 MW in the IEEE-39 system. Both the no-memory and full-memory controllers achieve strong suppression (~ 0.01), indicating that local softening dominates stabilization at this operating point.

III. RESULTS

A. Suppression of collapse probability

Figure 1 shows collapse probability as a function of injected load.

At 140 MW, the baseline collapse probability is approximately 0.56, while the controlled system reduces this to approximately 0.01, corresponding to $\Delta \approx 0.55$.

This raises the question of whether the observed suppression arises primarily from local softening or from the inclusion of memory feedback.

B. Ablation at a representative operating point

At this operating point, logistic softening alone already provides strong stabilization, such that the additional contribution of memory feedback is not statistically distinguishable within the observed variance.

IV. DISCUSSION

The results demonstrate that logistic softening can strongly suppress cascading failure in a benchmark system. While memory feedback is included in the control formulation, its additional contribution is not resolved at 140 MW.

Future work will include systematic exploration of parameter sensitivity, including variations in softening steepness, memory strength, and noise amplitude, as well as extension to larger systems where stabilization may be more marginal.

V. CONCLUSION

We have demonstrated that logistic softening with memory feedback can strongly suppress cascading failure in the IEEE-39 system. This provides a clean proof-of-concept and motivates further investigation in larger and more complex networks.